Production MLP Fitting Results

$\Delta P^{0.5}$ Stage-3 anchor-off five seeds and seed42 trajectory diagnostics

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Current Model

Production model

This deck now reports the true production MLP: Stage-3 `anchor_off`, seeds 7/17/42/99/2024, warm-started from the Stage-2 winner and refined on raw CDF supervision.

Inputs and scaling

The current variant is `a_dp050_plus_pressures`. Inputs include normalized time, tilt, plume count, injection duration, control backpressure, and log injection/chamber pressure. The target is still scaled with

$$A = (\Delta P)^{0.5} \rho_a^{-0.25} d_n^{0.5}.$$

Reading boundary

The five-seed mean is the thesis result. Seed42 is the lowest-RMSE production MLP seed and is used only for diagnostic plots.

Full-Clean Result

Model	RMSE	MAE	Bias	P95
Production MLP mean(5)	4.735 ± 0.037	3.293 ± 0.039	+0.626	9.167
Production MLP seed42	4.689	3.251	+0.486	9.092
SVGP Stage-3 seed42	4.628	3.173	-0.070	9.242

Model	Cov1	Cov2	mean std
Production MLP mean(5)	0.867	0.985	6.049 mm
Production MLP seed42	0.870	0.986	6.032 mm
SVGP Stage-3 seed42	0.703	0.939	4.000 mm

The careful conclusion: MLP does not beat SVGP on RMSE/MAE, but its uncertainty coverage is substantially more conservative.

Source: Thesis/generated/baseline_comparison_20260521/headline_full_clean.csv

Evaluation Size

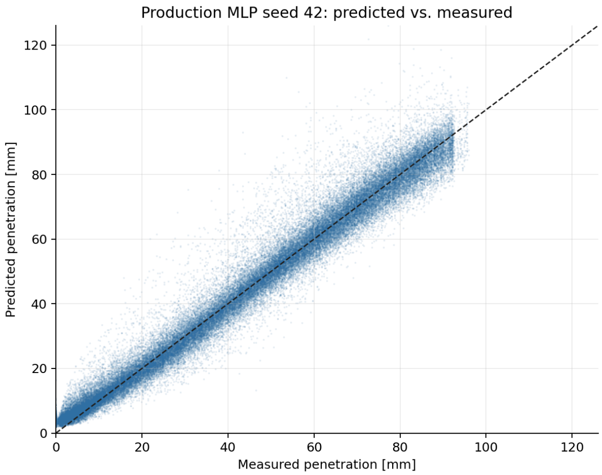
Item	Value
CSV files	2,754
Trajectories	24 316
Prediction points	692 942
Truth range	0.00 mm to 95.88 mm
Time window	0 ms to 5 ms

Representative seed42

- RMSE = 4.689 mm
- MAE = 3.251 mm
- Bias = +0.486 mm
- P95 absolute error = 9.092 mm
- $1\sigma/2\sigma$ coverage = 0.870 / 0.986

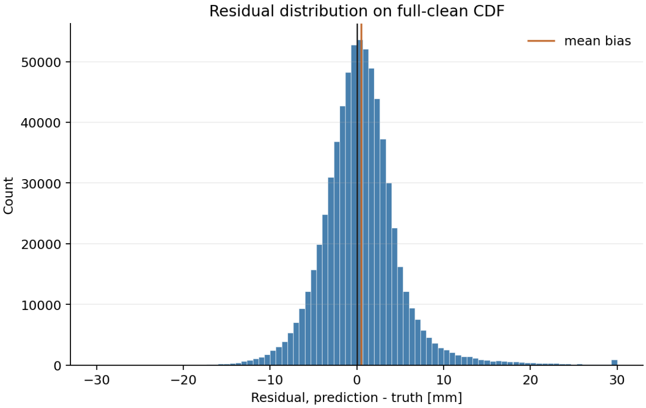
Source: MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/metrics_summary.json

Predicted vs. Measured Penetration



Source: Thesis/images/neural_network_fit_results_20260521/pred_vs_actual_seed42.png

Residual Distribution

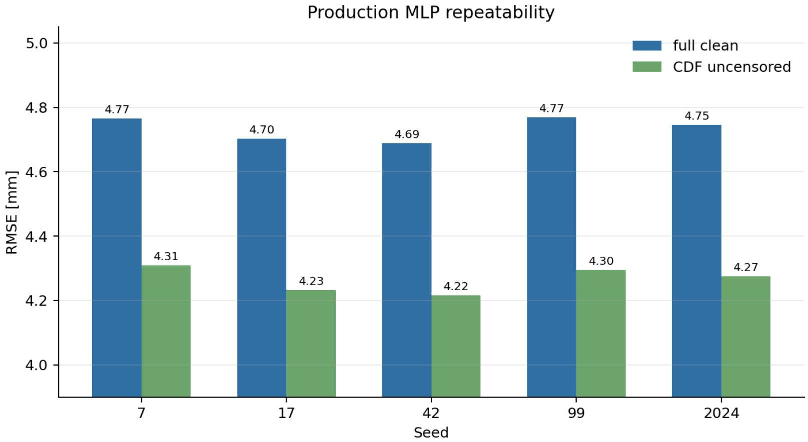


Residual definition

$$e = \hat{y}_{\text{pen}} - y_{\text{pen}}.$$

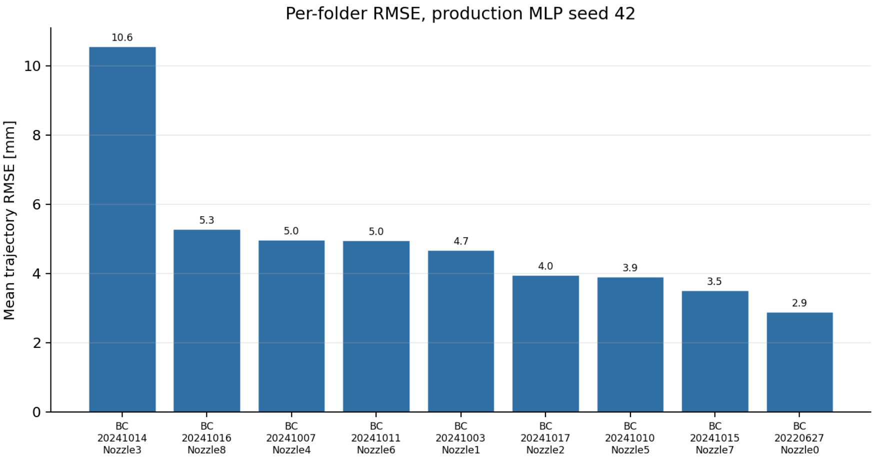
- seed42 bias is $+0.486\,$mm, small relative to RMSE.
- P95 absolute error is $9.092\,$mm.
- 2σ coverage is 0.986, above the Gaussian nominal 0.954.

Five-Seed Repeatability



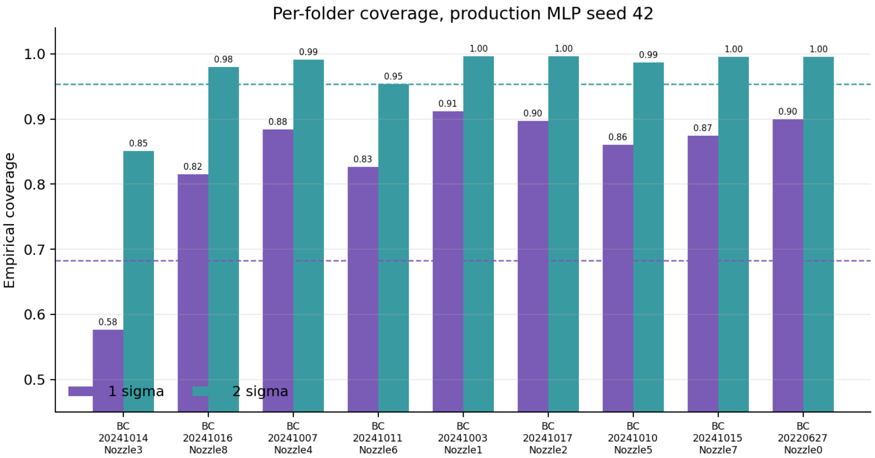
Source: Thesis/images/neural_network_fit_results_20260521/production_mlp_per_seed_rmse.png

Per-Folder RMSE



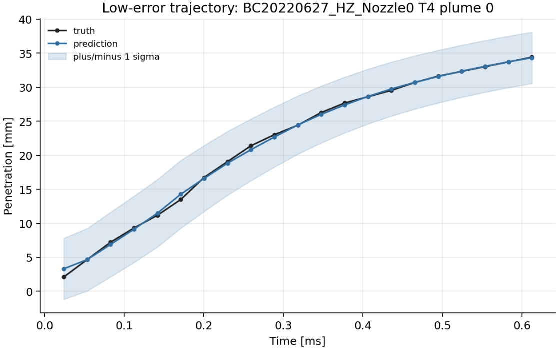
Source: MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/per_folder.csv

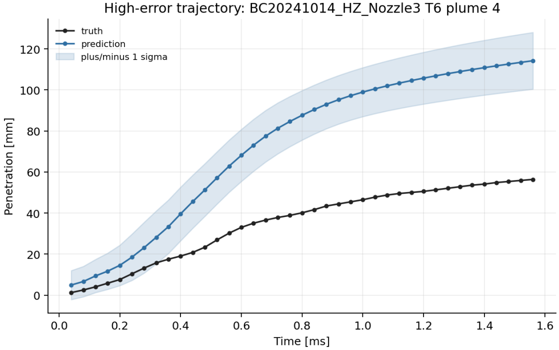
Per-Folder Coverage



Source: MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/per_folder.csv

Representative Trajectories





Nozzle0 T4 plume 0: RMSE 0.382 mm, coverage 1.00 / 1.00.

Nozzle3 T6 plume 4: RMSE 42.763 mm, coverage 0.205 / 0.282.

Source: Thesis/generated/neural_network_fit_results_20260521/summary.json

Residual Risk

1. Overall fitting is stable: five-seed full-clean RMSE is 4.735 ± 0.037 mm.
2. Uncertainty is conservative: this is acceptable for a screening surrogate.
3. Nozzle3 remains the main failure mode: it has the highest per-folder RMSE and weak worst-trajectory coverage.
4. Do not claim a full MLP win over SVGP: SVGP still has slightly better full-clean RMSE/MAE.

Thesis phrasing

Production MLP closes the gap to SVGP while providing more conservative uncertainty coverage, and it clearly outperforms HA/NS empirical curve baselines.

File Index

Item	File
Slides source	Thesis/slides/neural_network_fit_results_slides_en/neural_network_fit_results_slides_en.tex
Figures	Thesis/images/neural_network_fit_results_20260521/
Figure script	Thesis/generated/neural_network_fit_results_20260521/make_figures.py
Representative seed eval	MLP/eval/rmse_eval_clean_20260521_123603_distill_cdf_onset_v2_ablate_anchor_off_20260521_123126/
5-seed aggregate	Thesis/generated/baseline_comparison_20260521/production_mlp_full_clean_per_seed.csv